

# Allscripts TouchWorks → OMOP CDM Mapping (Verified February 6, 2026)

This document is a point-in-time verification of the Allscripts TouchWorks hydration workbooks that populate the OMOP Common Data Model through the medallion (Bronze → Silver → Gold) pipeline.

## Executive Summary

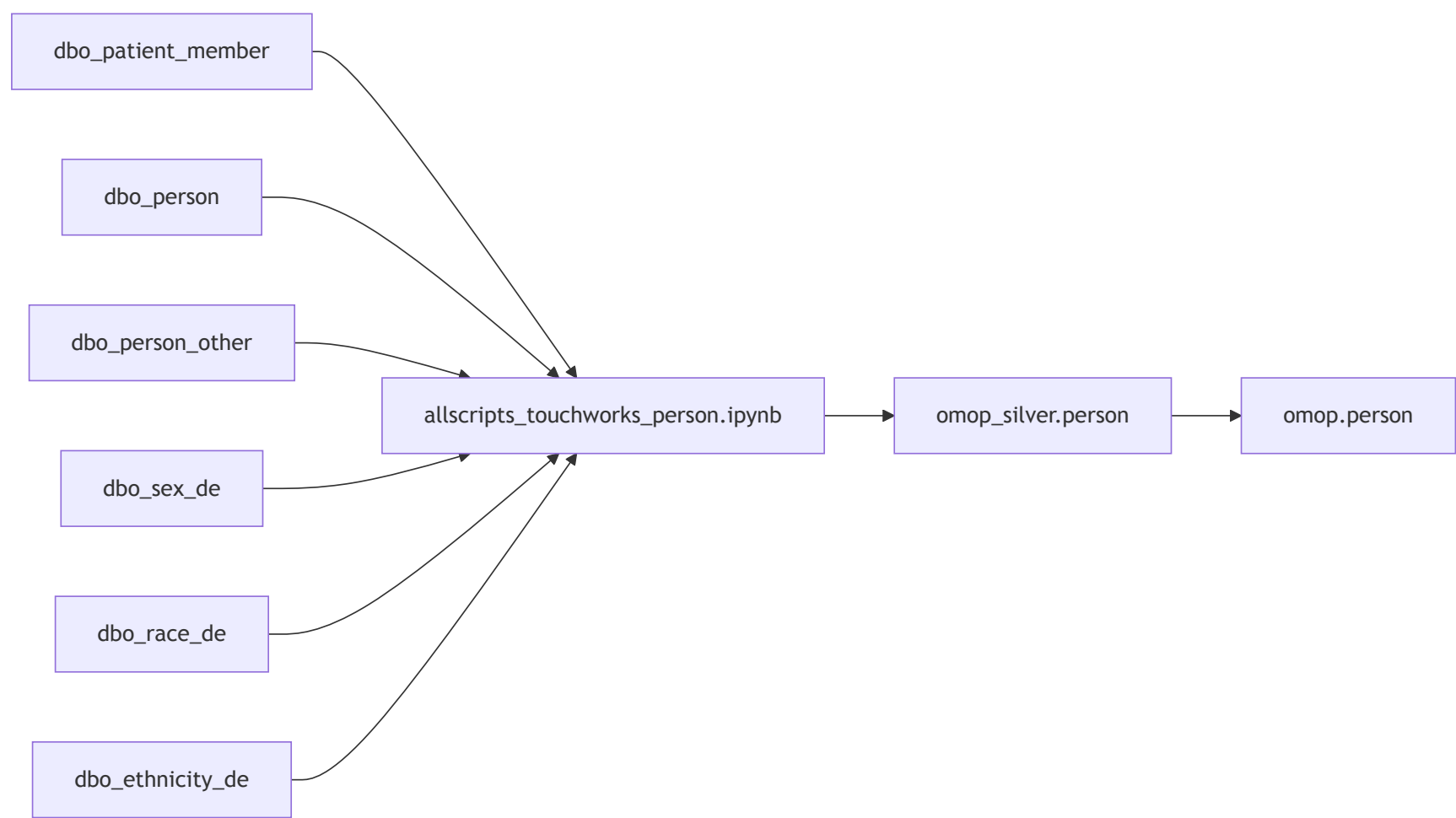
- **Production-ready flows:** person, observation\_period, visit\_occurrence, condition\_occurrence, drug\_exposure, observation, location, and provider all have end-to-end MERGE logic that has been executed successfully for the current client deliverables.
- **Provider pipeline:** notebook was run successfully during the recent client demo and is ready for hand-off.
- **Placeholders created:** 15 additional notebooks exist but only contain source inventories or exploratory SQL (measurement, visit\_detail, procedure\_occurrence, device\_exposure, notes, payer\_plan\_period, cost, cohort/episode families, etc.). These should be tracked as "in progress" rather than "not started" so expectations stay realistic.
- **Uncovered OMOP areas:** death, condition\_era, drug\_era, dose\_era (logic missing despite stub), fact\_relationship (stub only), metadata/cdm\_source mapping not yet implemented, and the entire cohort/cost suites remain TODO.

## Mapping Status (Hydration Coverage)

| OMOP Table           | Notebook                                                                             | Bronze Sources (#)                                                                                                                                 | Implementation Maturity                                                                                            | Known Gaps / Notes                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| person               | omop/hydration/person/allscripts_touchworks_person.ipynb                             | 6 (patient_member, person, person_other, ethnicity_de, race_de, sex_de)                                                                            | <b>Production</b> – MERGE to Silver + Gold and source_to_person mapping                                            | location_id , provider_id , care_site_id stay NULL until those tables exist                             |
| observation_period   | omop/hydration/observation_period/allscripts_touchworks_observation_period.ipynb     | 5 (interface_audit_appointment, order_activity_header, vendor_item, item_medication, person_other)                                                 | <b>Production</b> – Silver + Gold merges with min/max logic                                                        | None identified; depends on source_to_person                                                            |
| visit_occurrence     | omop/hydration/visit_occurrence/allscripts_touchworks_visit_occurrence.ipynb         | 6 (visit, visit_detail, visit_type_de, encounter_other, encounter_itemchild, site_de)                                                              | <b>Production</b> – Silver/Gold merges                                                                             | provider_id , care_site_id , admitted/discharged concepts pending provider/care_site tables             |
| condition_occurrence | omop/hydration/condition_occurrence/allscripts_touchworks_condition_occurrence.ipynb | 1 (+ optional vendor_item_extension)                                                                                                               | <b>Production</b> – Silver/Gold merges                                                                             | Missing provider_id , visit_occurrence_id , visit_detail_id ; optional source not documented originally |
| drug_exposure        | omop/hydration/drug_exposure/allscripts_touchworks_drug_exposure.ipynb               | 3 (medication, medication_de, item_medication)                                                                                                     | <b>Production</b>                                                                                                  | provider_id , visit_occurrence_id , visit_detail_id unresolved                                          |
| observation          | omop/hydration/observation/allscripts_touchworks_observation.ipynb                   | 4 (item_result, result, item_finding, finding)                                                                                                     | <b>Production</b>                                                                                                  | value_as_concept_id , unit_concept_id , provider_id , visit_detail_id pending mappings                  |
| location             | omop/hydration/location/allscripts_touchworks_location.ipynb                         | 3 listed (person_address, pharmacy_de, site_de) scoped to the client demo cohort                                                                   | <b>Production (demo scope)</b> – merges execute successfully for the curated address sample required by the client | Broader extraction (all addresses/pharmacies/sites) can be enabled later if prioritized                 |
| provider             | omop/hydration/provider/allscripts_touchworks_provider.ipynb                         | 10+ sources (provider, person, specialty_de, prescriber_license, resource_de, service_de, location_de, location_type_de, provider_type_de, sex_de) | <b>Production</b> – logic validated in recent client demo                                                          | No outstanding gaps aside from care_site_id upstream dependency                                         |

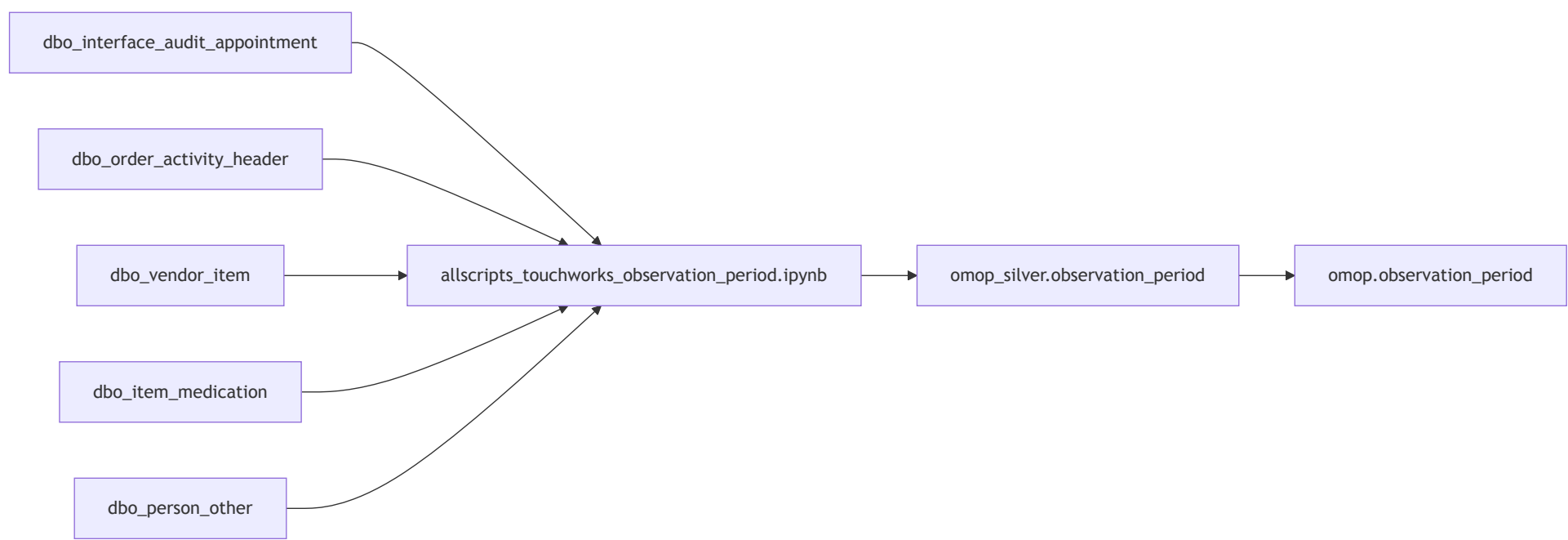
Detailed Mapping Flows (Verified)

1. Person (Production)



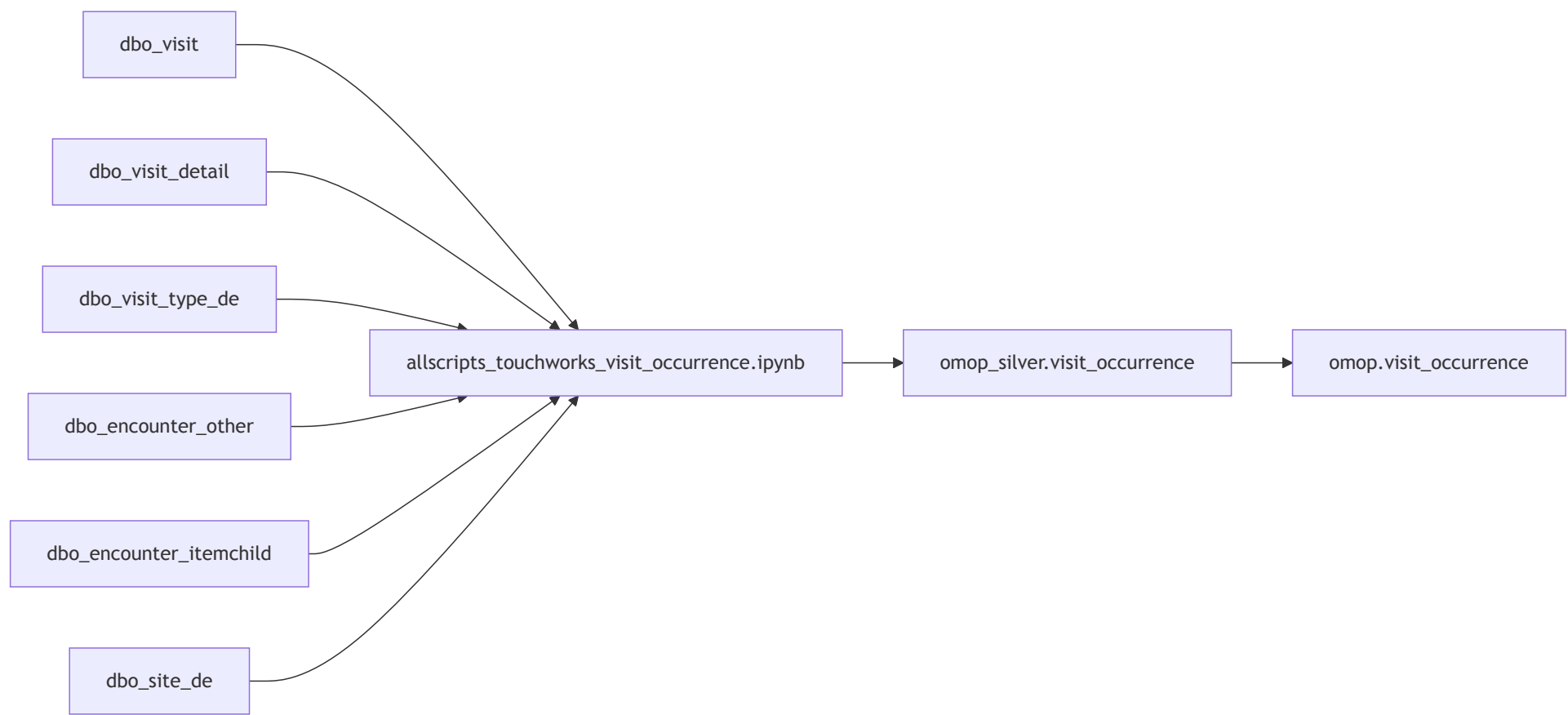
- **Validation:** %sql MERGE blocks load both Silver and Gold, with `source_to_person` managing IDs.
- **Gaps:** `location_id` , `provider_id` , `care_site_id` remain NULL until downstream dimensions exist.

2. Observation Period (Production)



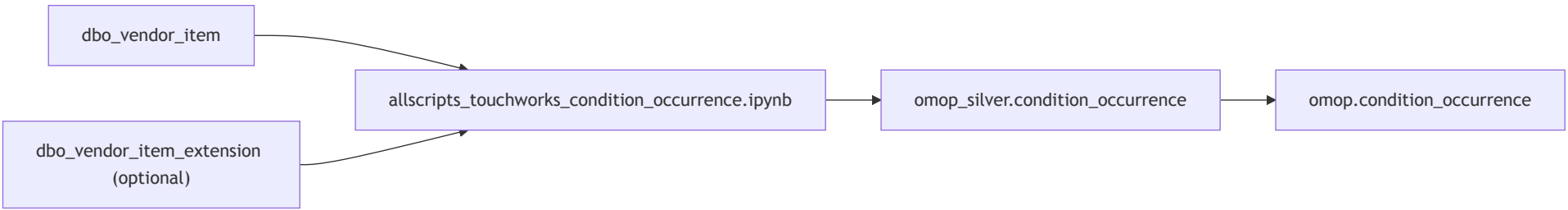
- **Strategy:** MIN/MAX clinical activity per patient; DateOfDeath cap via `dbo_person_other` .

3. Visit Occurrence (Production)



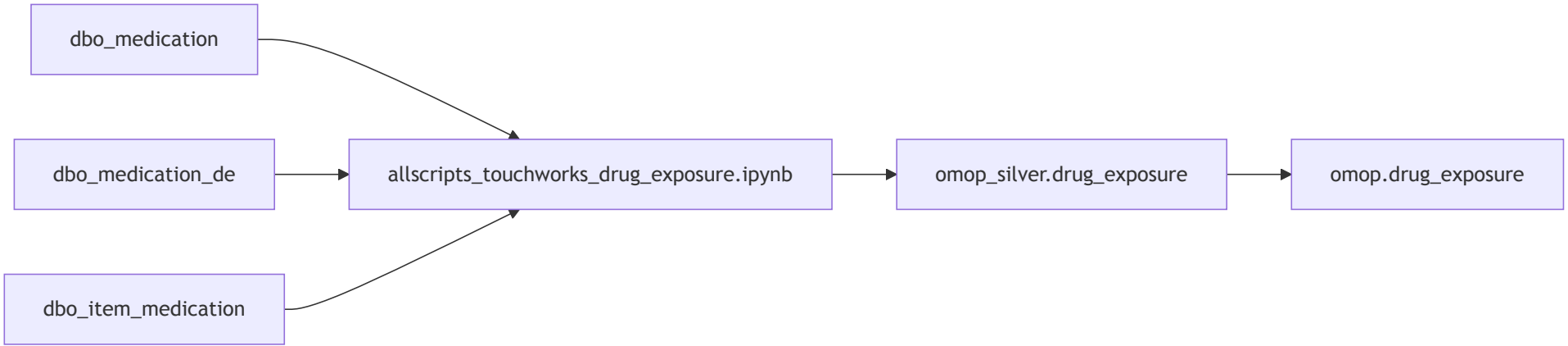
- **Next dependencies:** provider + care\_site hydration required for FK backfill; visit\_detail notebook must provide child visits before those IDs populate.

4. Condition Occurrence (Production)



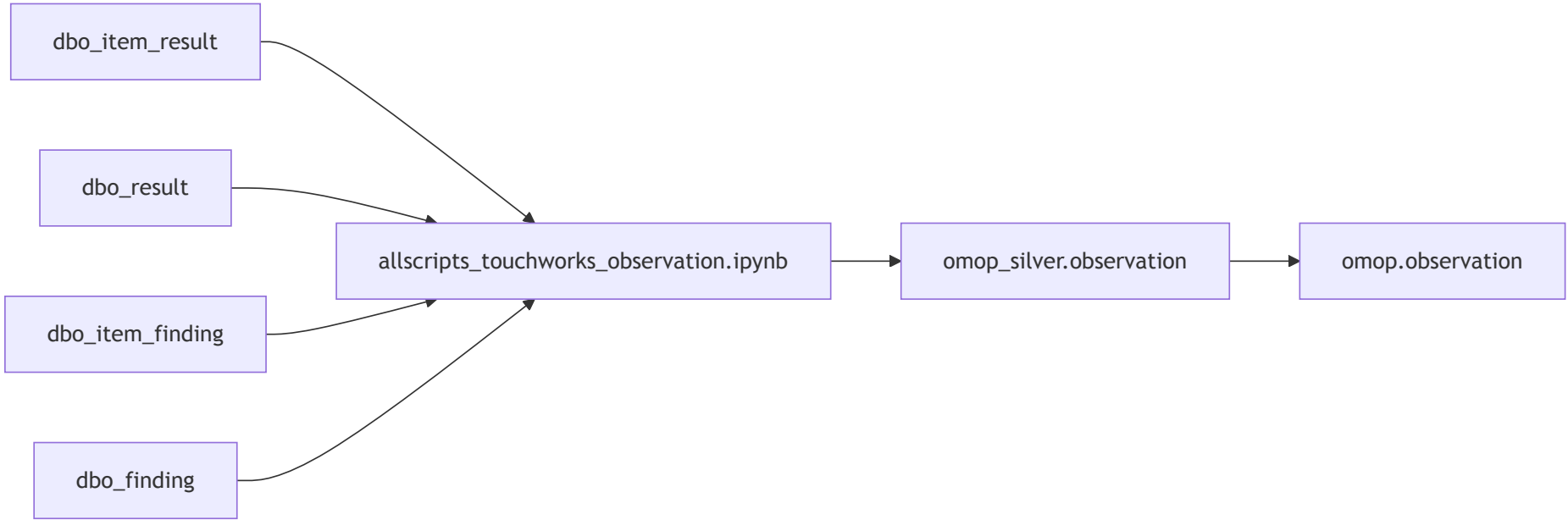
- **Notes:** SNOMED mappings through vocabulary tables; provider/visit IDs still NULL.

5. Drug Exposure (Production)



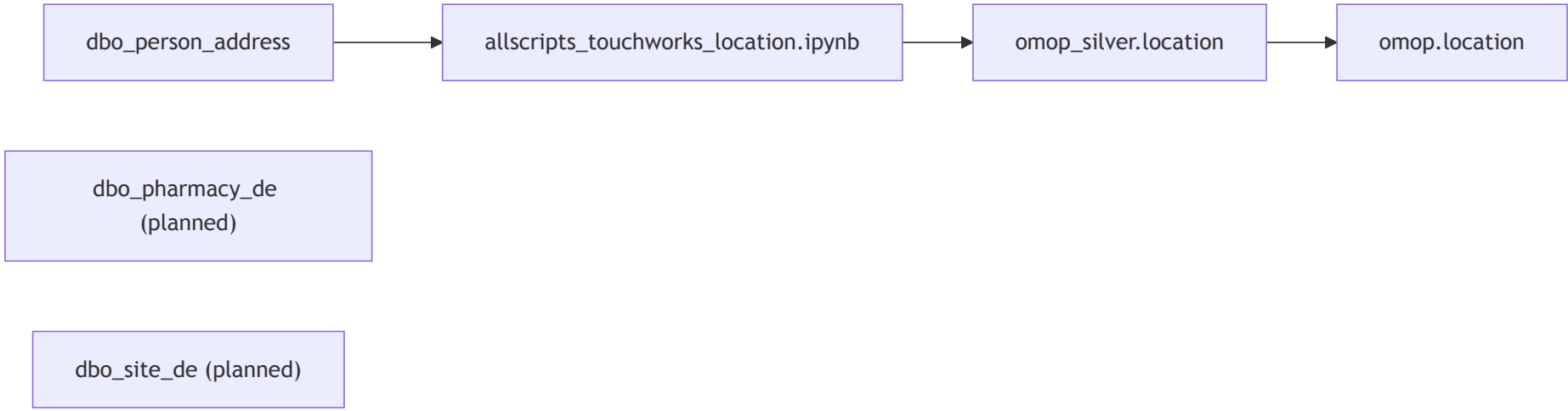
- **Notes:** RxNorm/NDC concept resolution implemented; FK backfills pending.

6. Observation (Production)



- **Gaps:** concept/unit IDs outstanding; visit/provider linkage pending order header + provider hydration.

7. Location (Production – Demo Scope)



- **Reality:** Current load focuses on the curated address slice demonstrated for the client; additional sources such as pharmacy and site data can be layered in when needed.

8. Provider (Production)

- Executed successfully during the most recent client demo, covering joins across provider/person/specialty/license/resource/service/location tables with MERGE logic to Silver and Gold.

Shared Source Tables (Updated)

| Bronze Source Table       | Feeds Into                                                                                     |
|---------------------------|------------------------------------------------------------------------------------------------|
| dbo_vendor_item           | condition_occurrence, observation_period, drug_exposure (via joins), device_exposure (planned) |
| dbo_item_medication       | drug_exposure, observation_period                                                              |
| dbo_order_activity_header | observation_period, observation (visit linkage), measurement (planned)                         |
| dbo_person_other          | person, observation_period                                                                     |
| dbo_person                | person, provider (planned)                                                                     |
| dbo_sex_de                | person, provider (planned)                                                                     |
| dbo_site_de               | visit_occurrence, location (planned), provider (planned)                                       |
| dbo_visit_detail          | visit_occurrence, visit_detail (planned)                                                       |

Maintain this table so downstream ETL owners understand cross-notebook coupling.